Batch Distillation

Final Lab Report
Unit Operations Lab 4
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1. Executive Summary (10 pts)

The objective of this experiment was to characterize the performance of a batch distillation column with an ethanol-water solution in order to quantify the relationship between boil-up rate and column hydrodynamics, the achievable degree of separation at given reflux conditions, and the overall column efficiency relative to ideal equilibrium staging. The results were used to determine scale-up and design decisions for similar binary distillation.

At total reflux, increasing the power from 0.75 to 1.50 kW increased the pressure drop from 74 Pa to 152 Pa and drove the column from a gentle foaming to violent foaming and bottom flooding. The distillate compositions that were measured under these conditions ranged from 65-84 mol% ethanol, depending on the boil up rate. Using the Fenske equation under total reflux gave seven theoretical stages, therefore corresponding to a 90.5% efficiency.

Under the finite reflux conditions, two reflux ratios were tested. The average distillate compositions were 83.1 mol% and 75.1 mole% ethanol. The corresponding bottoms compositions were 4.2 mol% amd 3.7 mol%, respectively. The theoretical stage counts were determined to be 5.8 and 4.1 and yielded column efficiencies of 72.5% and 51.3%.

Given these results, the column did not behave ideally at higher boil up rates and hydrodynamic limits like foaming and flooding degraded the separation and efficiency. Nevertheless, the column achieved high separation and reasonable efficiency under a controlled reflux and the measured performance was consistent with expected deviations from ideal equilibrium staging in industrial practice.

2. Introduction (10 pts)

Distillation is a widely used separation process in chemical industries due to its ability to handle diverse feeds and consistently achieve high-purity products (ChE 382 Lab Manuel, 2024). In both large-scale refining and fine chemical production, many facilities appear dominated by distillation columns, which reflects the centrality of phase equilibrium based separations to modern process design. When a liquid-phase mixture requires separation in a design problem, distillation is the default option in industry.

The objective of this experiment was to examine the behavior and performance of a binary ethanol-water distillation column. The process was operated under two conditions: (i) total reflux, to study boil up effects and column efficiency, and (ii) finite reflux operation, to observe changes in tops and bottoms compositions over time. The experiment was significant because it linked measurable operating variables such as boil up rate, temperature profile, pressure drop, and product composition to theoretical models of equilibrium staged separations. It also provided a practical basis for evaluating column performance using the Fenske equation, Murphee efficiency, and McCabe-Thiele graphical design methods.

The apparatus was an Armfield UOP3CC continuous distillation unit with a reboiler, condenser, reflux controller, sieve trays, and a control console with temperature and flow instrumentation (ChE 382 Lab Manual, 2024). The vapor that was generated in the reboiler rose counter currently against a descending reflux, contacting staged trays. The product vapor was then condensed and depending on the reflux setting, it returned to the column or was withdrawn as distillate. The tray temperatures, pressure drop, and compositions were monitored to characterize operation.

Industrial applications of binary and multiocomponent distillation span petroleum fractionation, solvent recovery, alcohol purification, flavor and fragrance processing, and fine chemical manufacturing. The methods applied in this experiment like Mcabe-Thiele stepping, total reflux benchmarking, and efficiency evaluation are directly compatible with industrial design packages used in commercial column sizing, revamp, and optimization.

3. Theory (10 pts)

Distillation can be defined as an equilibrium staged separation where a repeated vapor-liquid contact approaches phase equilibrium at each stage (ChE 382 Lab Manual, 2024). For a binary system, the McCabe-Thiele method provides a graphical solution by combining the VLE equilibrium curve, operating mass balance lines, and stepwise construction to count the required number of equilibrium stages (McCabe-Thiele Slides, 2024). A constant molal overflow and negligible heat losses were assumed.

All condensed vapor is returned to the column at total reflux; this condition minimizes the number of theoretical stages and enables benchmarking via the Fenske equation. In a binary system, the minimum number of theoretical plates is given by (ChE 382 Lab Manual, 2024):

$$n+1 \;=\; \frac{\log\left[\left(\frac{x_{EtOH}}{x_{H_2O}}\right)_d \left(\frac{x_{H_2O}}{x_{EtOH}}\right)_b\right]}{\log\left(\alpha_{EtOH/H_2O}\right)_{\text{avg}}}$$

Practical trays are not at full equilibrium, so the overall column efficiency is:

$$E = \frac{N_{\text{theoretical}}}{N_{\text{actual}}} \times 100\%$$

In a finite reflux, the McCabe-Thiele method uses observed top and bottom compositions and a chosen reflux ratio to draw operating lines and step off stages (McCabe-Thiele Slides, 2024). The number of equilibrium stages can then be compared to actual trays to extract efficiency.

The measured tray temperatures were mapped to composition via ethanol-water VLE. Distillate and bottoms compositions were entered in the Fenske and McCabe-Thiele frameworks. Boil up rate and pressure drop trends provided insight into hydrodynamics and tray behavior. These measurements connected the theoretical stage analysis to real column performance.

The experiment validated staged equilibrium distillation theory against real performance and enabled the extraction of design-relevant performance metrics using established distillation correlations.

4. Results (15 pts)

Table 1: Part A, pressure drop and tray observations.

Power (kW)	Pressure Drop	Pressure Difference	Degree of Foaming on Trays
0.75	70 to 144	74	Foaming gently over the whole tray
1	68 to 147	79	Foaming violently over the whole tray
1.25	53 to 163	110	Foaming violenty for 75% of the column
1.5	32 to 184	152	Violent foaming for top half of column; Liquid flooding in bottom half of column

Table 2: Part A, tops mole% ethanol, at different reboiler powers.

Power (kW)	Tops mol% ethanol
0.75	84.07
1	77.23
1.24	65.46
1.5	81.84

Table 3: Part A, Fenske equation for theoretical stages at 0.75kW, and efficiency.

Fenske Equation	Efficiency
7	90.48%

Table 4: Part B, tops and bottoms average mol% ethanol at different reflux rates, theoretical plates from the McCabe-Thiele method, and efficiencies.

	Reflux Rate 1	Reflux Rate 2
Tops Ethanol Ave. mol%	83.10	75.13
Bottoms Ethanol Ave. mol%	4.20	3.70
Theoretical Plates	5.8	4.1
Efficiency	72.50%	51.25%

5. Discussion (15 pts)

Generally, the results of the experiment match theoretical calculations and literature values, although a complete analysis of the distillation system could not be completed due to time and equipment restraints.

Data from experiment A, displayed in **table #**, indicates that the column consistently distills the ethanol from the water in solution at a range of different reboiler powers at total reflux. However, the efficacy of this separation varied, with 65.46% – 84.07% ethanol in the distillate and consistently low concentrations in the bottom product. Theoretically, the pressure drop across the column should linearly change with the vapor velocity. In experiment A, the pressure drop increases from 74 cm H_2O at 0.75kW to 152 cm H_2O at 1.50kW. The change in the pressure drop as the power increases (shown in **table #**) does not represent this linear relationship. The discrepancy between the experimental and theoretical values can most likely be attributed to the efficiency of the trays and the orifice-like behavior they exhibit once foaming, as these effects are assumed to be negligible, when they have an impact on the performance of the column. Utilizing the Fenske equation, approximately 7 theoretical plates were predicted (with 8 actual trays), somewhat validating the assumptions made in the experiment. This calculation also yielded a column efficiency of ~90.5%, although the calculation could not be completed at multiple reboiler powers, limiting the usefulness of this information. However, taking this efficiency to be an accurate average across the trials, this falls within the literature values for the efficiency of a sieve-tray distillation column of this size being used for an ethanol-water separation, further validating the results of experiment A [5]. The visual observations of the column in **table #**, that indicate that foaming increased as boil-up rate are also consistent with behavior mentioned in the literature, providing further proof that the column was operating correctly and lending credibility to the Fenske calculations [1]. Slight deviations from the expected values can be attributed to the same factors that affected the pressure drop.

In experiment B, using a 5:1 reflux ratio yielded a higher efficiency (72.50%) than the 10:1 ratio (51.25%), and the 5:1 ratio also had a higher concentration of ethanol in the distillate (83.10 mol% vs. 75.13 mol%). Both had very low (below 5 mol%) ethanol concentrations in the bottoms product. These findings align with the McCabe-Thiele calculations in **Appendix I**, which show a higher number of effective stages for the 5:1 ratio as opposed to the 10:1 ratio, due to a shift in the position of the operating line closer to the vapor-liquid equilibrium curve as the reflux ratio increases [2]. The 5:1 ratio is much more advantageous than the 10:1 ratio, as it provides a higher purity distillate product with a lower energy cost than the alternative.

Overall, the lack of data prevents definite conclusions being made, as several systematic errors need to be considered. These include heat loss over the non-insulated sections of the column, error in measuring the composition of the top and bottom products, and potential inconsistency when reading top and bottom collection. However, the experimental data's overall adherence to theoretical and literature values allow for more speculative conclusions to be drawn. The experimental data shows a correlation between boil-up rate and pressure drop and indicates that the reflux ratio has a significant impact on the separation efficiency of a column.

A major consideration for the results of these experiments is the effect of scope on the findings. Testing more reflux ratios and higher reboiler power rates would allow for a more thorough analysis of the system.

6. References (10 pts)

ChE 382: Unit Operations Laboratory — *Binary Batch Distillation*, Revision Fall 2024 v1, University of Illinois Chicago, 2024.

Fisher Scientific. (n.d.). https://www.fishersci.com/us/en/scientific-products/selection-guides/chemicals/ethanol.html

Google. (n.d.). *WO2000044697A1 - separation of ethanol and water from mixtures thereof by extractive distillation*. Google Patents. https://patents.google.com/patent/WO2000044697A1/en#

Liquid-vapour equilibrium data for ethanol-water at atmospheric pressure | download table. (n.d.). https://www.researchgate.net/figure/Liquid-vapour-equilibrium-data-for-ethanol-water-at-atmospheric-pressure_tbl1_272686708

McCabe–Thiele Graphical Design Method (Lecture Slides), School of Electrical Engineering Systems, DIT Kevin St., 2024.

7. Appendix I: Data Tabulation/Graphs (10 pts)

Table 5: Data from Part A of the lab, from the different powers used for the reboilers (0.75 to 1.5 kW).

		0.75			1			1.25			1.5		
		%v/v	%w/w	%mol/mol	%v/v	%w/w	%mol/mol	%v/v	%w/w	%mol/mol	%v/v	%w/w	%mol/mol
	1	94.14	92.69	83.21	91.66	89.66	77.23	86.00	82.90	65.46	93.59	92.01	81.84
	2	94.93	93.66	85.25									
Top	3	94.35	92.95	83.75									
	1	12.96	10.51	4.39									
	2	13.44	10.91	4.57									
Bottom	3	12.25	9.92	4.13									

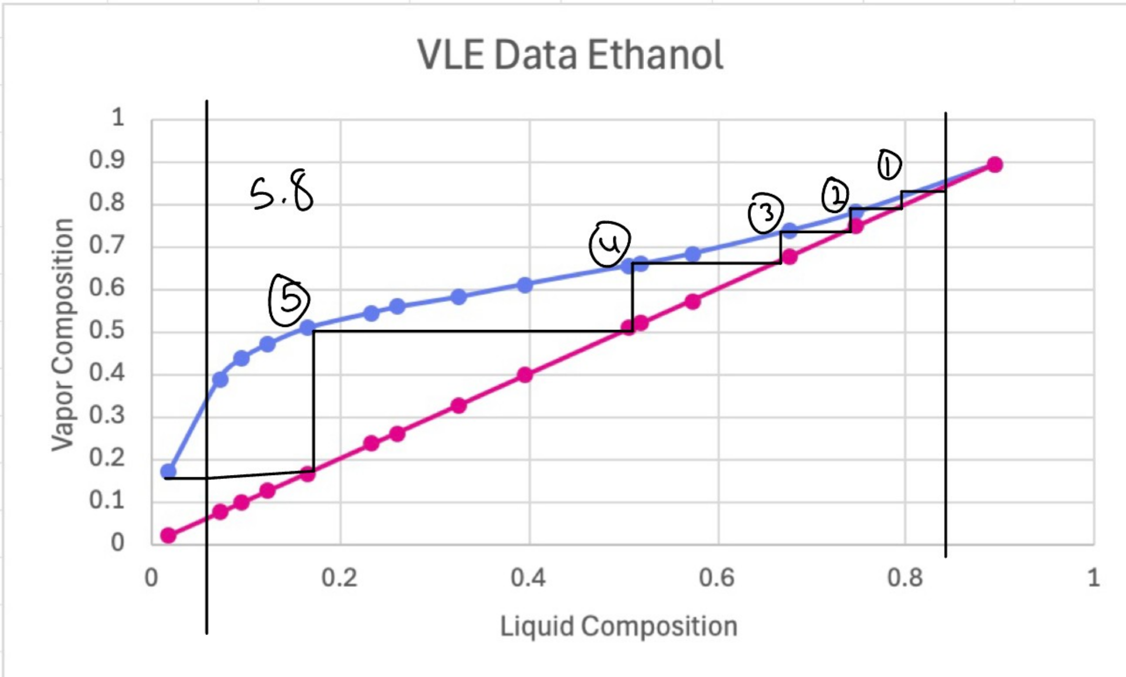
Table 6: Data from Part B, reflux rate one.

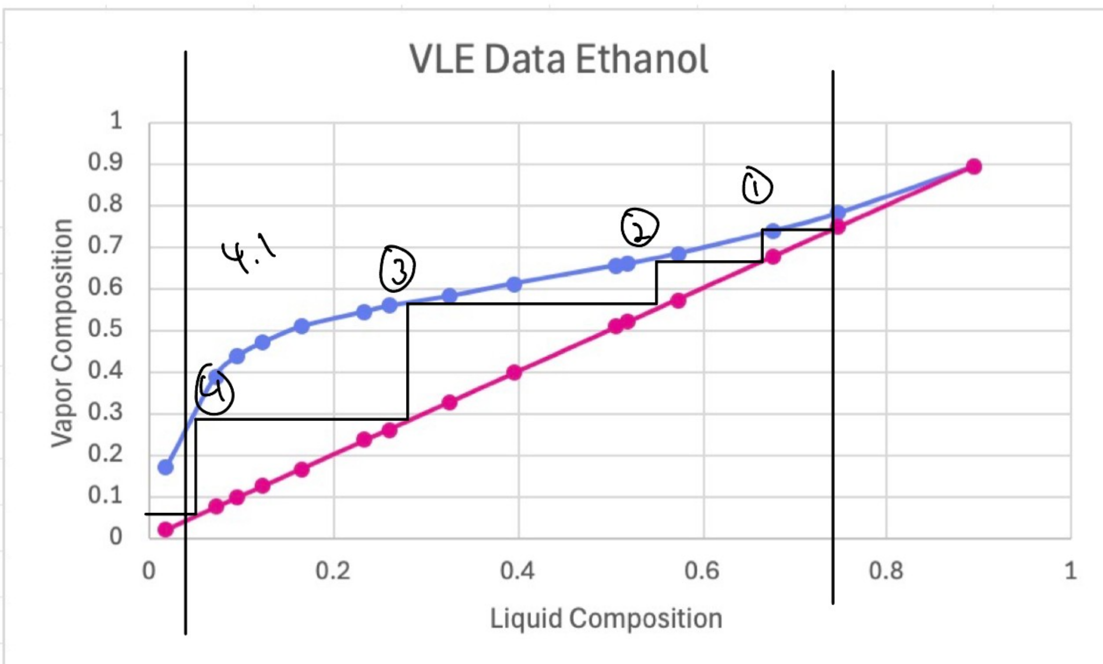
		RR 1		
		%v/v	%w/w	%mol/mol
Top	1	94.41	93.02	83.90
	2	94.22	92.79	83.42
	3	93.65	92.09	81.98
Bottom	1	12.6	10.21	4.26
	2	13.39	10.87	4.55
	3	11.31	9.14	3.79

Table 7: Data from Part B, reflux rate two.

		RR 2		
		%v/v	%w/w	%mol/mol
Top	1	90.05	87.72	73.63
	2	92.07	90.16	78.18
	3	90.02	87.68	73.57
Bottom	1	10.64	8.59	3.54
	2	11.9	9.63	4.00
	3	10.66	8.60	3.55

Graph(s) 1 and 2: In order of reflux rate experiments one and two, for Part B of the experiment.





8. Appendix II: Error Analysis (10 pts)

This experiment was affected by multiple sources of random and systematic error, which impacted both the accuracy and the precision of the results.

Random error in the system propagated through the measurement of the distillate and bottom compositions, as small fluctuations in the temperature, pressure over the column, and boil-up rate throughout each trial created random variation in the readings.

This error was compounded with random variation in the equipment readings due to limited measurement markings on the graduated cylinders used to create the initial ethanol-water solution and the manometer attached to the column.

Systematic errors were also present in the experiments through the thermocouples, densitometer, manometer, evaporation lose, hydrodynamic effects, power supply instability, and time measurement. The thermocouples used throughout the distillation column had an uncertainty of ± 0.5°C, which could impact the equilibrium assumption used in the calculations. The Snap 41 densitometer had calibration offsets between 0.5-1.0% of the readings. Additionally, this device utilized data at 20°C in its calculations, which may have been slightly different than the actual sample temperature. These factors may have affected the sample composition data. Aside from introducing some random error, the U-tube manometer utilized to measure the pressure drop had an error of ±0.2 cm H_2O. Evaporation lose, particularly when sampling the hot bottoms, may have caused lower readings for the bottoms product composition. Foaming, limited vapor-liquid contact within the column, and other hydrodynamic effects may have reduced the efficiency of the system in comparison to theoretical values. Additionally, small inconsistencies in the power supply may have prevented the system from reaching a completely steady state, affecting the equilibrium assumption used throughout the experiment. Some systematic error may have also been introduced through slight inconsistencies in the time measurement for sample collection

The effect of this error can be seen through the discrepancy between the theoretical and actual efficiency of the column (~90.5%), and the deviation from linear behavior observed in the change in pressure drop. Random error may have caused general scattering of the data, which can be quantified using the standard deviation:

$$s=\sqrt{\frac{\sum(x_i-x)^2}{n-1}}$$

Where s represents how much an individual measurement, x_i differs from the mean, x, over n trials. Given the limited number of trials conducted, this value is expected to be relatively large for this set of experiments. The systematic error mainly affects the values of the bottoms product ethanol concentration measurement and the pressure drop data, with each likely being somewhat suppressed.

Considering these sources of error, the data still matches expected values (from the theory and literature) well enough to be used to prove the general correlations discussed above, although definitive conclusions cannot be made.

9. Appendix III: Sample Calculations (10 pts)

Boil-up Rate would be the volumetric flow rate of the distillate divided by the volumetric flow rate of the bottoms. This would be found by dividing the volume collected within a certain amount of time converted into units of Liters/hr.

For a reboiler power of 0.75kW, and the first tops product collected and tested for ethanol v%/v, value 94.14, the following calculation to get to weight% was done.

$$\frac{\frac{94.14*0.789\,kg}{L}}{\frac{94.14*0.789\,kg}{L}+\frac{(100-94.14)*1\,kg}{L}}*100=92.69\,weight\,\%\,ethanol$$

And the following calculation was done to get mole%,

$$\frac{\frac{92.69*1\,kmole}{46.068\,kg}}{\frac{92.69*1\,kmole}{46.068\,kg}+\frac{(100-92.69)*1\,kmole}{18.016\,kg}}*100=83.21\,mol\,\%\,Ethanol$$

The Fenske equation was used to determine theoretical stages based on the average ethanol composition of the tops and bottoms. The average function was used in excel. The average tops mol% was 84% or 0.84, and the average bottoms mol% was 4% or 0.04. An alpha value of 1.8 was used due to literature values, found in the patent "Separation of ethanol and water from mixtures thereof by extractive distillation," stating a range between 1.1 and 2.4

$$n+1=\frac{\log\left(\left(\frac{0.84}{1-0.84}\right)*\left(\frac{1-0.04}{0.04}\right)\right)}{\log(1.8)}$$

Where n = 7.

Efficiency was calculated using the number of theoretical plates, 7, divided by the actual number of plates, 8, – for this case it would be 7/8, or approx. 90%.

The same conversion of v%/v to mole% was used for part B calcualtions, as well as efficiency, however the theoretical plates were found using the McCabe-Thiele method.

10. Appendix IV: Individual Team Contributions

NAME: Kendall

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Between lab set up, and the length of the experiment. It took a long time to heat the reboiler.
Pre-Lab Editing	0.5	Edited pre-lab as needed.
Experimental	0	

Objective (Prelab)		
Apparatus (prelab)	1	Took a photo of both the apparatus and the corresponding diagram, color coded labeling system.
Materials & Supplies (prelab)	0.5	Listed supplies needed for this lab.
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0.5	Looked up the safety data sheet and analyzed potential hazards for the lab – specifically heat and ethanol.
Final Lab Editing	1	Adjusted table and graph labels as needed.
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	5	Went over the data collected and analyzed it in the context of this lab.
Discussion (final lab)	0	
References (final and/or prelab)	0.5	Added sources used for VLE data.
Data Tabulation/Graphs (final)	2	Graphed VLE data, did the McCabe-Thiele.
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	2	Showed how I calculated the lab values from the collected raw data.
Power Point	1	Added the

Presentation		apparatus/materials supplies data, as referenced in the pre-lab. Adjusted according to feedback.
Total	20	

NAME: Lindsey

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	Spent the entire lab period on day 2 performing the experiment and spent 1 hour on day 1 gathering information
Pre-Lab Editing	2	Wrote out the procedure and safety evaluation and made sure everything sounded clear and concise
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	2	Outlined a step by step procedure using the lab manual and direction from the instructor
Project Safety Evaluation (prelab)	0.5	Evaluated safety hazards and ways to avoid them
Final Lab Editing	2	
Executive Summary (final lab)	1	Evaluated results and wrote out the expectation of the lab

		and what our results ended up being.
Introduction (Final lab)	1	Wrote what to expect in this lab and background on batch distillation
Theory (Final Lab)	1	Explained the equations and methods used in this lab
Results (final lab)	0	
Discussion (final lab)	0	
References (final and/or prelab)	0.25	Formatted and inserted references
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	0	
Power Point Presentation	1	Transferred information from Introduction and Theory to the presenation
Total	16.75	

NAME: Basil

	TIME (HOURS)	**DESCRIPTION**
Operator (Both Lab Days)	5	Making the ethanol-water solution and helping run the distillation column.
Pre-Lab Editing	0	
Experimental Objective (Prelab)	1.5	Wrote experimental objective and experimental intro.

Apparatus (prelab)	0	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	0	
Final Lab Editing	0.5	
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	5	Researching and writing discussion portion of the lab report.
References (final and/or prelab)	0.5	Added sources used in discussion section.
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	1.5	Completed error analysis calculations and write-up.
Sample Calculations (final Lab)	0	
Power Point Presentation	0.25	
Total	14.25	